

SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	The organization:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1a	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	Develops, documents, and disseminates to [Assignment: organization-defined personnel or roles]:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1a.1	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	A system and communications protection policy that addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1a.2	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	Procedures to facilitate the implementation of the system and communications protection policy and associated system and communications protection controls; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1b	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	Reviews and updates the current:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1b.1	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	System and communications protection policy [Assignment: organization-defined frequency]; and

SYSTEM AND COMMUNICATIONS PROTECTION	SC-1	SC-1b.2	SYSTEM AND COMMUNICATIONS PROTECTION POLICY AND PROCEDURES	System and communications protection procedures [Assignment: organization-defined frequency].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-2	SC-2	APPLICATION PARTITIONING	The information system separates user functionality (including user interface services) from information system management functionality.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-2 (1)	SC-2 (1)	INTERFACES FOR NON-PRIVILEGED USERS	The information system prevents the presentation of information system management-related functionality at an interface for non-privileged users.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-3	SC-3	SECURITY FUNCTION ISOLATION	The information system isolates security functions from nonsecurity functions.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-3 (1)	SC-3 (1)	HARDWARE SEPARATION	The information system utilizes underlying hardware separation mechanisms to implement security function isolation.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-3 (2)	SC-3 (2)	ACCESS / FLOW CONTROL FUNCTIONS	The information system isolates security functions enforcing access and information flow control from nonsecurity functions and from other security functions.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-3 (3)	SC-3 (3)	MINIMIZE NONSECURITY FUNCTIONALITY	The organization minimizes the number of nonsecurity functions included within the isolation boundary containing security functions.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-3 (4)	SC-3 (4)	MODULE COUPLING AND COHESIVENESS	The organization implements security functions as largely independent modules that maximize internal cohesiveness within modules and minimize coupling between modules.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-3 (5)	SC-3 (5)	LAYERED STRUCTURES	The organization implements security functions as a layered structure minimizing interactions between layers of the design and avoiding any dependence by lower layers on the functionality or correctness of higher layers.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-4	SC-4	INFORMATION IN SHARED RESOURCES	The information system prevents unauthorized and unintended information transfer via shared system resources.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-4 (1)	SC-4 (1)	SECURITY LEVELS	[Withdrawn: Incorporated into SC-4].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-4 (2)	SC-4 (2)	PERIODS PROCESSING	The information system prevents unauthorized information transfer via shared resources in accordance with [Assignment: organization-defined procedures] when system processing explicitly switches between different information classification levels or security categories.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-5	SC-5	DENIAL OF SERVICE PROTECTION	The information system protects against or limits the effects of the following types of denial of service attacks: [Assignment: organization-defined types of denial of service attacks or references to sources for such information] by employing [Assignment: organization-defined security safeguards].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-5 (1)	SC-5 (1)	RESTRICT INTERNAL USERS	The information system restricts the ability of individuals to launch [Assignment: organization-defined denial of service attacks] against other information systems.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-5 (2)	SC-5 (2)	EXCESS CAPACITY / BANDWIDTH / REDUNDANCY	The information system manages excess capacity, bandwidth, or other redundancy to limit the effects of information flooding denial of service attacks.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-5 (3)	SC-5 (3)	DETECTION / MONITORING	The organization:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-5 (3)	SC-5 (3)(a)	DETECTION / MONITORING	Employs [Assignment: organization-defined monitoring tools] to detect indicators of denial of service attacks against the information system; and

SYSTEM AND COMMUNICATIONS PROTECTION	SC-5 (3)	SC-5 (3)(b)	DETECTION / MONITORING	Monitors [Assignment: organization-defined information system resources] to determine if sufficient resources exist to prevent effective denial of service attacks.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-6	SC-6	RESOURCE AVAILABILITY	The information system protects the availability of resources by allocating [Assignment: organization-defined resources] by [Selection (one or more); priority; quota; [Assignment: organization-defined security safeguards]].

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-7

SC-7

BOUNDARY
PROTECTION

The information system:

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-7

SC-7a

BOUNDARY
PROTECTION

Monitors and controls communications at the
external boundary of the system and at key
internal boundaries within the system;

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-7

SC-7b

BOUNDARY
PROTECTION

Implements subnetworks for publicly accessible system components that are [Selection: physically; logically] separated from internal organizational networks; and

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7	SC-7c	BOUNDARY PROTECTION	Connects to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (1)	SC-7 (1)	PHYSICALLY SEPARATED SUBNETWORKS	[Withdrawn: Incorporated into SC-7].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (2)	SC-7 (2)	PUBLIC ACCESS	[Withdrawn: Incorporated into SC-7].

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (3)	SC-7 (3)	ACCESS POINTS	The organization limits the number of external network connections to the information system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (4)	SC-7 (4)	EXTERNAL TELECOMMUNICATIONS SERVICES	The organization:

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (4)	SC-7 (4)(a)	EXTERNAL TELECOMMUNICATIONS SERVICES	Implements a managed interface for each external telecommunication service;
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SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-7 (4)

SC-7 (4)(b)

EXTERNAL
TELECOMMUNICATIO
NS SERVICES

Establishes a traffic flow policy for each
managed interface;

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (4)	SC-7 (4)(c)	EXTERNAL TELECOMMUNICATIONS SERVICES	Protects the confidentiality and integrity of the information being transmitted across each interface;
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SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (4)	SC-7 (4)(d)	EXTERNAL TELECOMMUNICATIONS SERVICES	Documents each exception to the traffic flow policy with a supporting mission/business need and duration of that need; and
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SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (4)	SC-7 (4)(e)	EXTERNAL TELECOMMUNICATIONS SERVICES	Reviews exceptions to the traffic flow policy [Assignment: organization-defined frequency] and removes exceptions that are no longer supported by an explicit mission/business need.
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SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (5)	SC-7 (5)	DENY BY DEFAULT / ALLOW BY EXCEPTION	The information system at managed interfaces denies network communications traffic by default and allows network communications traffic by exception (i.e., deny all, permit by exception).
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (6)	SC-7 (6)	RESPONSE TO RECOGNIZED FAILURES	[Withdrawn: Incorporated into SC-7 (18)].

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-7 (7)

SC-7 (7)

PREVENT SPLIT
TUNNELING FOR
REMOTE DEVICES

The information system, in conjunction with a remote device, prevents the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (8)	SC-7 (8)	ROUTE TRAFFIC TO AUTHENTICATED PROXY SERVERS	The information system routes [Assignment: organization-defined internal communications traffic] to [Assignment: organization-defined external networks] through authenticated proxy servers at managed interfaces.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (9)	SC-7 (9)	RESTRICT THREATENING OUTGOING COMMUNICATIONS TRAFFIC	The information system:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (9)	SC-7 (9)(a)	RESTRICT THREATENING OUTGOING COMMUNICATIONS TRAFFIC	Detects and denies outgoing communications traffic posing a threat to external information systems; and

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (9)	SC-7 (9)(b)	RESTRICT THREATENING OUTGOING COMMUNICATIONS TRAFFIC	Audits the identity of internal users associated with denied communications.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (10)	SC-7 (10)	PREVENT UNAUTHORIZED EXFILTRATION	The organization prevents the unauthorized exfiltration of information across managed interfaces.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (11)	SC-7 (11)	RESTRICT INCOMING COMMUNICATIONS TRAFFIC	The information system only allows incoming communications from [Assignment: organization-defined authorized sources] to be routed to [Assignment: organization-defined authorized destinations].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (12)	SC-7 (12)	HOST-BASED PROTECTION	The organization implements [Assignment: organization-defined host-based boundary protection mechanisms] at [Assignment: organization-defined information system components].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (13)	SC-7 (13)	ISOLATION OF SECURITY TOOLS / MECHANISMS / SUPPORT COMPONENTS	The organization isolates [Assignment: organization-defined information security tools, mechanisms, and support components] from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (14)	SC-7 (14)	PROTECTS AGAINST UNAUTHORIZED PHYSICAL CONNECTIONS	The organization protects against unauthorized physical connections at [Assignment: organization-defined managed interfaces].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (15)	SC-7 (15)	ROUTE PRIVILEGED NETWORK ACCESSES	The information system routes all networked, privileged accesses through a dedicated, managed interface for purposes of access control and auditing.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (16)	SC-7 (16)	PREVENT DISCOVERY OF COMPONENTS / DEVICES	The information system prevents discovery of specific system components composing a managed interface.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (17)	SC-7 (17)	AUTOMATED ENFORCEMENT OF PROTOCOL FORMATS	The information system enforces adherence to protocol formats.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (18)	SC-7 (18)	FAIL SECURE	The information system fails securely in the event of an operational failure of a boundary protection device.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (19)	SC-7 (19)	BLOCKS COMMUNICATION FROM NON-ORGANIZATIONALLY CONFIGURED HOSTS	The information system blocks both inbound and outbound communications traffic between [Assignment: organization-defined communication clients] that are independently configured by end users and external service providers.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (20)	SC-7 (20)	DYNAMIC ISOLATION / SEGREGATION	The information system provides the capability to dynamically isolate/segregate [Assignment: organization-defined information system components] from other components of the system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (21)	SC-7 (21)	ISOLATION OF INFORMATION SYSTEM COMPONENTS	The organization employs boundary protection mechanisms to separate [Assignment: organization-defined information system components] supporting [Assignment: organization-defined missions and/or business functions].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (22)	SC-7 (22)	SEPARATE SUBNETS FOR CONNECTING TO DIFFERENT SECURITY DOMAINS	The information system implements separate network addresses (i.e., different subnets) to connect to systems in different security domains.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-7 (23)	SC-7 (23)	DISABLE SENDER FEEDBACK ON PROTOCOL VALIDATION FAILURE	The information system disables feedback to senders on protocol format validation failure.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-8	SC-8	TRANSMISSION CONFIDENTIALITY AND INTEGRITY	The information system protects the [Selection (one or more): confidentiality; integrity] of transmitted information.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-8 (1)	SC-8 (1)	CRYPTOGRAPHIC OR ALTERNATE PHYSICAL PROTECTION	The information system implements cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission unless otherwise protected by [Assignment: organization-defined alternative physical safeguards].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-8 (2)	SC-8 (2)	PRE / POST TRANSMISSION HANDLING	The information system maintains the [Selection (one or more): confidentiality; integrity] of information during preparation for transmission and during reception.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-8 (3)	SC-8 (3)	CRYPTOGRAPHIC PROTECTION FOR MESSAGE EXTERNALS	The information system implements cryptographic mechanisms to protect message externals unless otherwise protected by [Assignment: organization-defined alternative physical safeguards].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-8 (4)	SC-8 (4)	CONCEAL / RANDOMIZE COMMUNICATIONS	The information system implements cryptographic mechanisms to conceal or randomize communication patterns unless otherwise protected by [Assignment: organization-defined alternative physical safeguards].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-9	SC-9	TRANSMISSION CONFIDENTIALITY	[Withdrawn: Incorporated into SC-8].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-10	SC-10	NETWORK DISCONNECT	The information system terminates the network connection associated with a communications session at the end of the session or after [Assignment: organization-defined time period] of inactivity.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-11	SC-11	TRUSTED PATH	The information system establishes a trusted communications path between the user and the following security functions of the system: [Assignment: organization-defined security functions to include at a minimum, information system authentication and re-authentication].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-11 (1)	SC-11 (1)	LOGICAL ISOLATION	The information system provides a trusted communications path that is logically isolated and distinguishable from other paths.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-12	SC-12	CRYPTOGRAPHIC KEY ESTABLISHMENT AND MANAGEMENT	The organization establishes and manages cryptographic keys for required cryptography employed within the information system in accordance with [Assignment: organization-defined requirements for key generation, distribution, storage, access, and destruction].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-12 (1)	SC-12 (1)	AVAILABILITY	The organization maintains availability of information in the event of the loss of cryptographic keys by users.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-12 (2)	SC-12 (2)	SYMMETRIC KEYS	The organization produces, controls, and distributes symmetric cryptographic keys using [Selection: NIST FIPS-compliant; NSA-approved] key management technology and processes.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-12 (3)	SC-12 (3)	ASYMMETRIC KEYS	The organization produces, controls, and distributes asymmetric cryptographic keys using [Selection: NSA-approved key management technology and processes; approved PKI Class 3 certificates or prepositioned keying material; approved PKI Class 3 or Class 4 certificates and hardware security tokens that protect the user's private key].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-12 (4)	SC-12 (4)	PKI CERTIFICATES	[Withdrawn: Incorporated into SC-12].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-12 (5)	SC-12 (5)	PKI CERTIFICATES / HARDWARE TOKENS	[Withdrawn: Incorporated into SC-12].

SYSTEM AND COMMUNICATIONS PROTECTION	SC-13	SC-13	CRYPTOGRAPHIC PROTECTION	The information system implements [Assignment: organization-defined cryptographic uses and type of cryptography required for each use] in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, and standards.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-13 (1)	SC-13 (1)	FIPS-VALIDATED CRYPTOGRAPHY	[Withdrawn: Incorporated into SC-13].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-13 (2)	SC-13 (2)	NSA-APPROVED CRYPTOGRAPHY	[Withdrawn: Incorporated into SC-13].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-13 (3)	SC-13 (3)	INDIVIDUALS WITHOUT FORMAL ACCESS APPROVALS	[Withdrawn: Incorporated into SC-13].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-13 (4)	SC-13 (4)	DIGITAL SIGNATURES	[Withdrawn: Incorporated into SC-13].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-14	SC-14	PUBLIC ACCESS PROTECTIONS	[Withdrawn: Capability provided by AC-2, AC-3, AC-5, AC-6, SI-3, SI-4, SI-5, SI-7, SI-10].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-15	SC-15	COLLABORATIVE COMPUTING DEVICES	The information system:

SYSTEM AND COMMUNICATIONS PROTECTION	SC-15	SC-15a	COLLABORATIVE COMPUTING DEVICES	Prohibits remote activation of collaborative computing devices with the following exceptions: [Assignment: organization-defined exceptions where remote activation is to be allowed]; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-15	SC-15b	COLLABORATIVE COMPUTING DEVICES	Provides an explicit indication of use to users physically present at the devices.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-15 (1)	SC-15 (1)	PHYSICAL DISCONNECT	The information system provides physical disconnect of collaborative computing devices in a manner that supports ease of use.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-15 (2)	SC-15 (2)	BLOCKING INBOUND / OUTBOUND COMMUNICATIONS TRAFFIC	[Withdrawn: Incorporated into SC-7].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-15 (3)	SC-15 (3)	DISABLING / REMOVAL IN SECURE WORK AREAS	The organization disables or removes collaborative computing devices from [Assignment: organization-defined information systems or information system components] in [Assignment: organization-defined secure work areas].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-15 (4)	SC-15 (4)	EXPLICITLY INDICATE CURRENT PARTICIPANTS	The information system provides an explicit indication of current participants in [Assignment: organization-defined online meetings and teleconferences].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-16	SC-16	TRANSMISSION OF SECURITY ATTRIBUTES	The information system associates [Assignment: organization-defined security attributes] with information exchanged between information systems and between system components.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-16 (1)	SC-16 (1)	INTEGRITY VALIDATION	The information system validates the integrity of transmitted security attributes.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-17	SC-17	PUBLIC KEY INFRASTRUCTURE CERTIFICATES	The organization issues public key certificates under an [Assignment: organization-defined certificate policy] or obtains public key certificates from an approved service provider.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18	SC-18	MOBILE CODE	The organization:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18	SC-18a	MOBILE CODE	Defines acceptable and unacceptable mobile code and mobile code technologies;
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18	SC-18b	MOBILE CODE	Establishes usage restrictions and implementation guidance for acceptable mobile code and mobile code technologies; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18	SC-18c	MOBILE CODE	Authorizes, monitors, and controls the use of mobile code within the information system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18 (1)	SC-18 (1)	IDENTIFY UNACCEPTABLE CODE / TAKE CORRECTIVE ACTIONS	The information system identifies [Assignment: organization-defined unacceptable mobile code] and takes [Assignment: organization-defined corrective actions].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18 (2)	SC-18 (2)	ACQUISITION / DEVELOPMENT / USE	The organization ensures that the acquisition, development, and use of mobile code to be deployed in the information system meets [Assignment: organization-defined mobile code requirements].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18 (3)	SC-18 (3)	PREVENT DOWNLOADING / EXECUTION	The information system prevents the download and execution of [Assignment: organization-defined unacceptable mobile code].

SYSTEM AND COMMUNICATIONS PROTECTION	SC-18 (4)	SC-18 (4)	PREVENT AUTOMATIC EXECUTION	The information system prevents the automatic execution of mobile code in [Assignment: organization-defined software applications] and enforces [Assignment: organization-defined actions] prior to executing the code.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-18 (5)	SC-18 (5)	ALLOW EXECUTION ONLY IN CONFINED ENVIRONMENTS	The organization allows execution of permitted mobile code only in confined virtual machine environments.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-19	SC-19	VOICE OVER INTERNET PROTOCOL	The organization:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-19	SC-19a	VOICE OVER INTERNET PROTOCOL	Establishes usage restrictions and implementation guidance for Voice over Internet Protocol (VoIP) technologies based on the potential to cause damage to the information system if used maliciously; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-19	SC-19b	VOICE OVER INTERNET PROTOCOL	Authorizes, monitors, and controls the use of VoIP within the information system.

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-20

SC-20

SECURE NAME /
ADDRESS
RESOLUTION SERVICE
(AUTHORITATIVE
SOURCE)

The information system:

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-20

SC-20a

SECURE NAME /
ADDRESS
RESOLUTION SERVICE
(AUTHORITATIVE
SOURCE)

Provides additional data origin authentication
and integrity verification artifacts along with the
authoritative name resolution data the system
returns in response to external name/address
resolution queries; and

SYSTEM AND COMMUNICATIONS PROTECTION	SC-20	SC-20b	SECURE NAME / ADDRESS RESOLUTION SERVICE (AUTHORITATIVE SOURCE)	Provides the means to indicate the security status of child zones and (if the child supports secure resolution services) to enable verification of a chain of trust among parent and child domains, when operating as part of a distributed, hierarchical namespace.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-20 (1)	SC-20 (1)	CHILD SUBSPACES	[Withdrawn: Incorporated into SC-20].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-20 (2)	SC-20 (2)	DATA ORIGIN / INTEGRITY	The information system provides data origin and integrity protection artifacts for internal name/address resolution queries.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-21	SC-21	SECURE NAME / ADDRESS RESOLUTION SERVICE (RECURSIVE OR CACHING RESOLVER)	The information system requests and performs data origin authentication and data integrity verification on the name/address resolution responses the system receives from authoritative sources.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-21 (1)	SC-21 (1)	DATA ORIGIN / INTEGRITY	[Withdrawn: Incorporated into SC-21].

SYSTEM AND
COMMUNICATIONS
PROTECTION

SC-22

SC-22

ARCHITECTURE AND
PROVISIONING FOR
NAME / ADDRESS
RESOLUTION SERVICE

The information systems that collectively
provide name/address resolution service for an
organization are fault-tolerant and implement
internal/external role separation.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-23	SC-23	SESSION AUTHENTICITY	The information system protects the authenticity of communications sessions.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-23 (1)	SC-23 (1)	INVALIDATE SESSION IDENTIFIERS AT LOGOUT	The information system invalidates session identifiers upon user logout or other session termination.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-23 (2)	SC-23 (2)	USER-INITIATED LOGOUTS / MESSAGE DISPLAYS	[Withdrawn: Incorporated into AC-12 (1)].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-23 (3)	SC-23 (3)	UNIQUE SESSION IDENTIFIERS WITH RANDOMIZATION	The information system generates a unique session identifier for each session with [Assignment: organization-defined randomness requirements] and recognizes only session identifiers that are system-generated.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-23 (4)	SC-23 (4)	UNIQUE SESSION IDENTIFIERS WITH RANDOMIZATION	[Withdrawn: Incorporated into SC-23 (3)].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-23 (5)	SC-23 (5)	ALLOWED CERTIFICATE AUTHORITIES	The information system only allows the use of [Assignment: organization-defined certificate authorities] for verification of the establishment of protected sessions.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-24	SC-24	FAIL IN KNOWN STATE	The information system fails to a [Assignment: organization-defined known-state] for [Assignment: organization-defined types of failures] preserving [Assignment: organization-defined system state information] in failure.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-25	SC-25	THIN NODES	The organization employs [Assignment: organization-defined information system components] with minimal functionality and information storage.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-26	SC-26	HONEYPOTS	The information system includes components specifically designed to be the target of malicious attacks for the purpose of detecting, deflecting, and analyzing such attacks.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-26 (1)	SC-26 (1)	DETECTION OF MALICIOUS CODE	[Withdrawn: Incorporated into SC-35].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-27	SC-27	PLATFORM-INDEPENDENT APPLICATIONS	The information system includes: [Assignment: organization-defined platform-independent applications].

SYSTEM AND COMMUNICATIONS PROTECTION	SC-28	SC-28	PROTECTION OF INFORMATION AT REST	The information system protects the [Selection (one or more): confidentiality; integrity] of [Assignment: organization-defined information at rest].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-28 (1)	SC-28 (1)	CRYPTOGRAPHIC PROTECTION	The information system implements cryptographic mechanisms to prevent unauthorized disclosure and modification of [Assignment: organization-defined information] on [Assignment: organization-defined information system components].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-28 (2)	SC-28 (2)	OFF-LINE STORAGE	The organization removes from online storage and stores off-line in a secure location [Assignment: organization-defined information].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-29	SC-29	HETEROGENEITY	The organization employs a diverse set of information technologies for [Assignment: organization-defined information system components] in the implementation of the information system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-29 (1)	SC-29 (1)	VIRTUALIZATION TECHNIQUES	The organization employs virtualization techniques to support the deployment of a diversity of operating systems and applications that are changed [Assignment: organization-defined frequency].

SYSTEM AND COMMUNICATIONS PROTECTION	SC-30	SC-30	CONCEALMENT AND MISDIRECTION	The organization employs [Assignment: organization-defined concealment and misdirection techniques] for [Assignment: organization-defined information systems] at [Assignment: organization-defined time periods] to confuse and mislead adversaries.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-30 (1)	SC-30 (1)	VIRTUALIZATION TECHNIQUES	[Withdrawn: Incorporated into SC-29 (1)].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-30 (2)	SC-30 (2)	RANDOMNESS	The organization employs [Assignment: organization-defined techniques] to introduce randomness into organizational operations and assets.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-30 (3)	SC-30 (3)	CHANGE PROCESSING / STORAGE LOCATIONS	The organization changes the location of [Assignment: organization-defined processing and/or storage] [Selection: [Assignment: organization-defined time frequency]; at random time intervals]].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-30 (4)	SC-30 (4)	MISLEADING INFORMATION	The organization employs realistic, but misleading information in [Assignment: organization-defined information system components] with regard to its security state or posture.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-30 (5)	SC-30 (5)	CONCEALMENT OF SYSTEM COMPONENTS	The organization employs [Assignment: organization-defined techniques] to hide or conceal [Assignment: organization-defined information system components].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-31	SC-31	COVERT CHANNEL ANALYSIS	The organization:

SYSTEM AND COMMUNICATIONS PROTECTION	SC-31	SC-31a	COVERT CHANNEL ANALYSIS	Performs a covert channel analysis to identify those aspects of communications within the information system that are potential avenues for covert [Selection (one or more): storage; timing] channels; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-31	SC-31b	COVERT CHANNEL ANALYSIS	Estimates the maximum bandwidth of those channels.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-31 (1)	SC-31 (1)	TEST COVERT CHANNELS FOR EXPLOITABILITY	The organization tests a subset of the identified covert channels to determine which channels are exploitable.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-31 (2)	SC-31 (2)	MAXIMUM BANDWIDTH	The organization reduces the maximum bandwidth for identified covert [Selection (one or more); storage; timing] channels to [Assignment: organization-defined values].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-31 (3)	SC-31 (3)	MEASURE BANDWIDTH IN OPERATIONAL ENVIRONMENTS	The organization measures the bandwidth of [Assignment: organization-defined subset of identified covert channels] in the operational environment of the information system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-32	SC-32	INFORMATION SYSTEM PARTITIONING	The organization partitions the information system into [Assignment: organization-defined information system components] residing in separate physical domains or environments based on [Assignment: organization-defined circumstances for physical separation of components].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-33	SC-33	TRANSMISSION PREPARATION INTEGRITY	[Withdrawn: Incorporated into SC-8].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34	SC-34	NON-MODIFIABLE EXECUTABLE PROGRAMS	The information system at [Assignment: organization-defined information system components]:

SYSTEM AND COMMUNICATIONS PROTECTION	SC-34	SC-34a	NON-MODIFIABLE EXECUTABLE PROGRAMS	Loads and executes the operating environment from hardware-enforced, read-only media; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34	SC-34b	NON-MODIFIABLE EXECUTABLE PROGRAMS	Loads and executes [Assignment: organization-defined applications] from hardware-enforced, read-only media.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34 (1)	SC-34 (1)	NO WRITABLE STORAGE	The organization employs [Assignment: organization-defined information system components] with no writeable storage that is persistent across component restart or power on/off.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34 (2)	SC-34 (2)	INTEGRITY PROTECTION / READ-ONLY MEDIA	The organization protects the integrity of information prior to storage on read-only media and controls the media after such information has been recorded onto the media.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34 (3)	SC-34 (3)	HARDWARE-BASED PROTECTION	The organization:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34 (3)	SC-34 (3)(a)	HARDWARE-BASED PROTECTION	Employs hardware-based, write-protect for [Assignment: organization-defined information system firmware components]; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-34 (3)	SC-34 (3)(b)	HARDWARE-BASED PROTECTION	Implements specific procedures for [Assignment: organization-defined authorized individuals] to manually disable hardware write-protect for firmware modifications and re-enable the write-protect prior to returning to operational mode.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-35	SC-35	HONEYCLIENTS	The information system includes components that proactively seek to identify malicious websites and/or web-based malicious code.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-36	SC-36	DISTRIBUTED PROCESSING AND STORAGE	The organization distributes [Assignment: organization-defined processing and storage] across multiple physical locations.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-36	SC-36	DISTRIBUTED PROCESSING AND STORAGE	The organization distributes [Assignment: organization-defined processing and storage] across multiple physical locations.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-36 (1)	SC-36 (1)	POLLING TECHNIQUES	The organization employs polling techniques to identify potential faults, errors, or compromises to [Assignment: organization-defined distributed processing and storage components].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-37	SC-37	OUT-OF-BAND CHANNELS	The organization employs [Assignment: organization-defined out-of-band channels] for the physical delivery or electronic transmission of [Assignment: organization-defined information, information system components, or devices] to [Assignment: organization-defined individuals or information systems].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-37 (1)	SC-37 (1)	ENSURE DELIVERY / TRANSMISSION	The organization employs [Assignment: organization-defined security safeguards] to ensure that only [Assignment: organization-defined individuals or information systems] receive the [Assignment: organization-defined information, information system components, or devices].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-38	SC-38	OPERATIONS SECURITY	The organization employs [Assignment: organization-defined operations security safeguards] to protect key organizational information throughout the system development life cycle.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-39	SC-39	PROCESS ISOLATION	The information system maintains a separate execution domain for each executing process.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-39 (1)	SC-39 (1)	HARDWARE SEPARATION	The information system implements underlying hardware separation mechanisms to facilitate process separation.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-39 (2)	SC-39 (2)	THREAD ISOLATION	The information system maintains a separate execution domain for each thread in [Assignment: organization-defined multi-threaded processing].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-40	SC-40	WIRELESS LINK PROTECTION	The information system protects external and internal [Assignment: organization-defined wireless links] from [Assignment: organization-defined types of signal parameter attacks or references to sources for such attacks].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-40 (1)	SC-40 (1)	ELECTROMAGNETIC INTERFERENCE	The information system implements cryptographic mechanisms that achieve [Assignment: organization-defined level of protection] against the effects of intentional electromagnetic interference.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-40 (2)	SC-40 (2)	REDUCE DETECTION POTENTIAL	The information system implements cryptographic mechanisms to reduce the detection potential of wireless links to [Assignment: organization-defined level of reduction].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-40 (3)	SC-40 (3)	IMITATIVE OR MANIPULATIVE COMMUNICATIONS DECEPTION	The information system implements cryptographic mechanisms to identify and reject wireless transmissions that are deliberate attempts to achieve imitative or manipulative communications deception based on signal parameters.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-40 (4)	SC-40 (4)	SIGNAL PARAMETER IDENTIFICATION	The information system implements cryptographic mechanisms to prevent the identification of [Assignment: organization-defined wireless transmitters] by using the transmitter signal parameters.

SYSTEM AND COMMUNICATIONS PROTECTION	SC-41	SC-41	PORT AND I/O DEVICE ACCESS	The organization physically disables or removes [Assignment: organization-defined connection ports or input/output devices] on [Assignment: organization-defined information systems or information system components].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-42	SC-42	SENSOR CAPABILITY AND DATA	The information system:
SYSTEM AND COMMUNICATIONS PROTECTION	SC-42	SC-42a	SENSOR CAPABILITY AND DATA	Prohibits the remote activation of environmental sensing capabilities with the following exceptions: [Assignment: organization-defined exceptions where remote activation of sensors is allowed]; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-42	SC-42b	SENSOR CAPABILITY AND DATA	Provides an explicit indication of sensor use to [Assignment: organization-defined class of users].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-42 (1)	SC-42 (1)	REPORTING TO AUTHORIZED INDIVIDUALS OR ROLES	The organization ensures that the information system is configured so that data or information collected by the [Assignment: organization-defined sensors] is only reported to authorized individuals or roles.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-42 (2)	SC-42 (2)	AUTHORIZED USE	The organization employs the following measures: [Assignment: organization-defined measures], so that data or information collected by [Assignment: organization-defined sensors] is only used for authorized purposes.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-42 (3)	SC-42 (3)	PROHIBIT USE OF DEVICES	The organization prohibits the use of devices possessing [Assignment: organization-defined environmental sensing capabilities] in [Assignment: organization-defined facilities, areas, or systems].
SYSTEM AND COMMUNICATIONS PROTECTION	SC-43	SC-43	USAGE RESTRICTIONS	The organization:

SYSTEM AND COMMUNICATIONS PROTECTION	SC-43	SC-43a	USAGE RESTRICTIONS	Establishes usage restrictions and implementation guidance for [Assignment: organization-defined information system components] based on the potential to cause damage to the information system if used maliciously; and
SYSTEM AND COMMUNICATIONS PROTECTION	SC-43	SC-43b	USAGE RESTRICTIONS	Authorizes, monitors, and controls the use of such components within the information system.
SYSTEM AND COMMUNICATIONS PROTECTION	SC-44	SC-44	DETONATION CHAMBERS	The organization employs a detonation chamber capability within [Assignment: organization-defined information system, system component, or location].
SECURITY	SE-1	SE-1	Inventory of Personally Identifiable Information	Privacy Control. Refer to NIST SP 800-53 (Rev. 4) Appendix J
SECURITY	SE-2	SE-2	Privacy Incident Response	Privacy Control. Refer to NIST SP 800-53 (Rev. 4) Appendix J
SYSTEM AND INFORMATION INTEGRITY	SI-1	SI-1	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	The organization:
SYSTEM AND INFORMATION INTEGRITY	SI-1	SI-1a	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	Develops, documents, and disseminates to [Assignment: organization-defined personnel or roles]:
SYSTEM AND INFORMATION INTEGRITY	SC-1	SI-1a.1	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	A system and information integrity policy that addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and

SYSTEM AND INFORMATION INTEGRITY	SC-1	SI-1a.2	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	Procedures to facilitate the implementation of the system and information integrity policy and associated system and information integrity controls; and
SYSTEM AND INFORMATION INTEGRITY	SI-1	SI-1b	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	Reviews and updates the current:
SYSTEM AND INFORMATION INTEGRITY	SC-1	SI-1b.1	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	System and information integrity policy [Assignment: organization-defined frequency]; and
SYSTEM AND INFORMATION INTEGRITY	SC-1	SI-1b.2	SYSTEM AND INFORMATION INTEGRITY POLICY AND PROCEDURES	System and information integrity procedures [Assignment: organization-defined frequency].